Ch 5.1.1-2: Leave One Out Cross-validation

Lecture 10 - CMSE 381

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Announcements UDPATE

Last time:

- ~~Lab on Discrimination~~

Classification

Announcements:

- Fourth homework due Friday! Covers:
  - Fri 2/4 Lab
  - Mon 2/7 Lecture
  - Weds 2/9 Lecture
- Office hours - ONLINE

Covered in this lecture

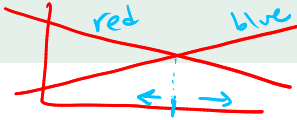
- LOO CV
- Outliers
- Leverage statistic

Section 1

Last time

LDA in one slide

$p=1$



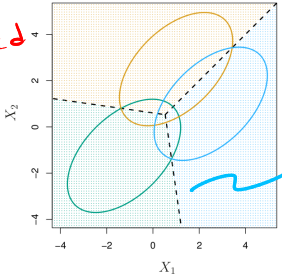
δ_{blue}

δ_{red}

- Assume observations in the kth class drawn from normal distribution with same covariance matrix for all k: $N(\mu_k, \Sigma)$
- For new data point x, predict the k for which $p_k(x) = \Pr(Y = k \mid X = x)$ is largest.
- Equivalent to finding k for which $\delta_k(x)$ is largest

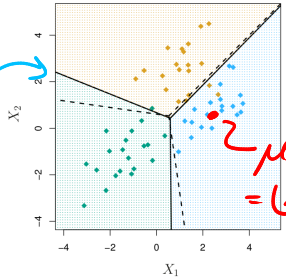
$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k$$

- Approximate μ_i's, π_i's, and Σ to find boundary where the returned k switches
- Setup means the boundaries are linear



Bayes

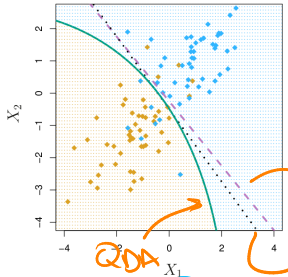
LDA

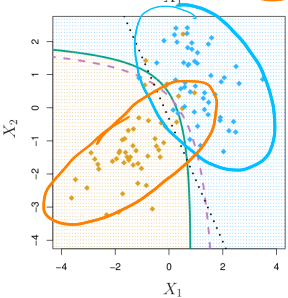


$\mu_{blue} = (2.5, 1)$

QDA in one slide

- Same idea as LDA, but don't assume same covariance matrix
- Assume observations in kth class drawn from normal distribution $N(\mu_k, \Sigma_k)$
- Setup means boundaries are quadratic





Types of errors

(x_i, y_i)

What happened

$n = \#$ data pts

		True class		
		− or Null	+ or Non-null	Total
Predicted class	− or Null	True Neg. (TN)	False Neg. (FN)	N^*
	+ or Non-null	False Pos. (FP)	True Pos. (TP)	P^*
	Total	N	P	

$(x_i, \hat{y}_i)$

$= N + P$

$= N^* + P^*$

Name	Definition	Synonyms
False Pos. rate	FP/N	Type I error, 1−Specificity
True Pos. rate	TP/P	1−Type II error, power, sensitivity, recall
Pos. Pred. value	TP/P^*	Precision, 1−false discovery proportion
Neg. Pred. value	TN/N^*	

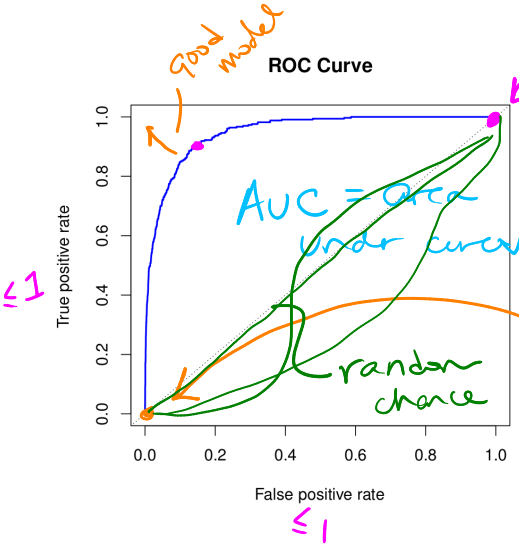
$\frac{FP}{N} \leq 1$

$\frac{TP}{P} \leq 1$

$1 - \text{Type II} = TP/P$

$1 - TP/P = \text{Type II}$

ROC curve



Every choice of threshold for the model $\Rightarrow (FPR, TPR)$

Classifier always predict +

	−	+	
−	0	0	o
+	a	b	n
	a	b	

$$\frac{FP}{N} = \frac{a}{a} = 1$$

$$\frac{TP}{P} = \frac{b}{b} = 1$$

Classifier always −

	−	+	
−	a	b	n
+	0	0	

$$\frac{FP}{N} = \frac{0}{a} = 0$$

$$\frac{TP}{P} = \frac{0}{b} = 0$$

Section 2

5.1 Validation set

What's the problem?

- How well is my ML method doing? *Model Assessment*
- Which method is best for our data?
- How many features should I use? Which ones? *Model selection*
- What is the uncertainty in the learned parameters?

This time: Leave one out cross-validation

LOOCV

Training Error vs Testing Error

Training Error

Calculated by applying stats learning method to the observations used in training (MSE)

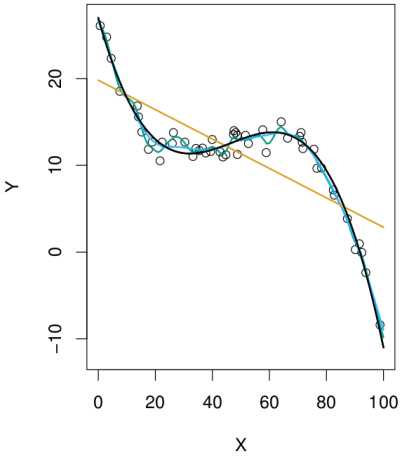
- Massive under estimate

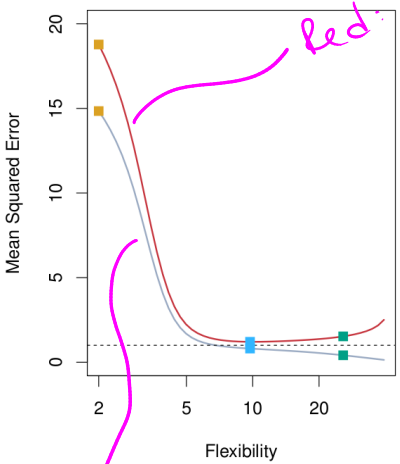
↳ overfitting!!!

Testing Error

Avg error that results from using model built to predict a new observation (not used in training)

Throw-back Wednesday

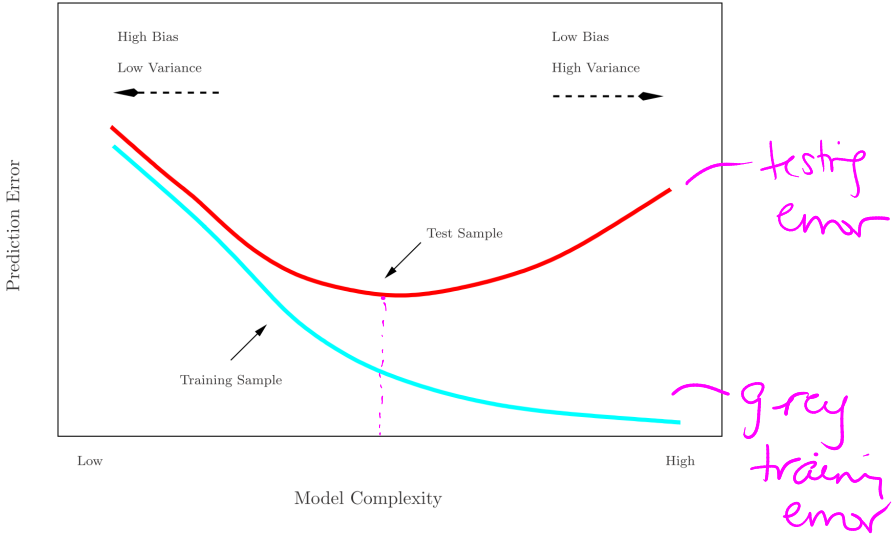




Red: testing error

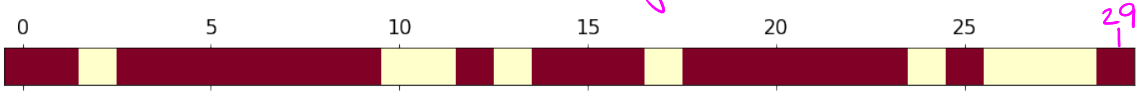
Gray: training error

Model tradeoffs



Validation set approach

Resampling 1.0



- Divide randomly into two parts:
  - Training set
  - Validation/Hold-out/Testing set
- Fit model on training set ~ Red
- Use fitted model to predict response for observations in the test set
- Evaluate quality (e.g. MSE)

21 pts

70%

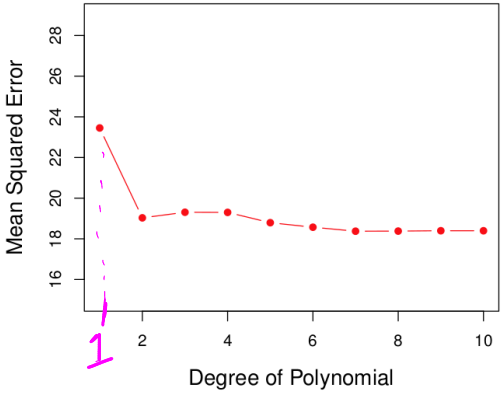
Ex) 30 data pts

30% of data

↳ yellow

testing set

Example with the auto data



Ex] Same test set for every polynomial

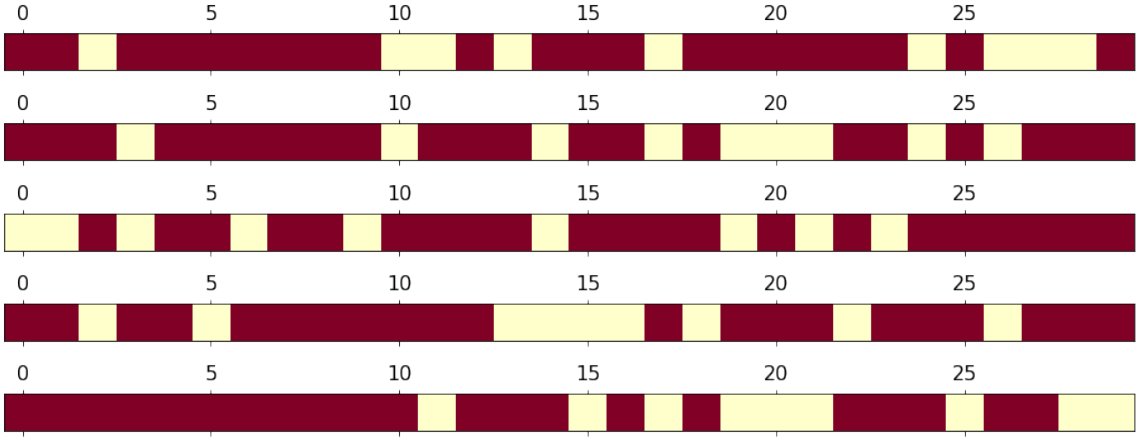
Predicting mpg using horsepower:

$$\texttt{mpg} = \boxed{\beta_0 + \beta_1 \texttt{hp} + \beta_2 \texttt{hp}^2} + \cdots + \beta_p \texttt{hp}^p$$

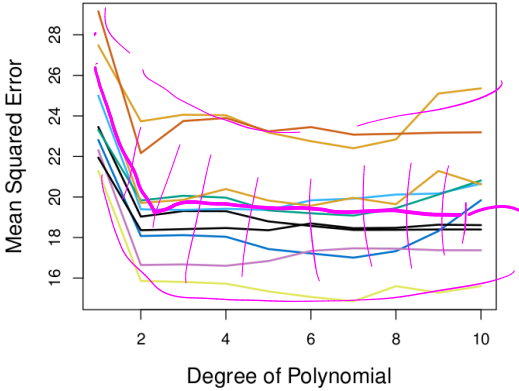
· Test error

· Conclude linear not good enough

Rinse and repeat



Again example with auto data



- Repeat procedure w/ different splits
- Highly variable results
- Stats models better w/ more data

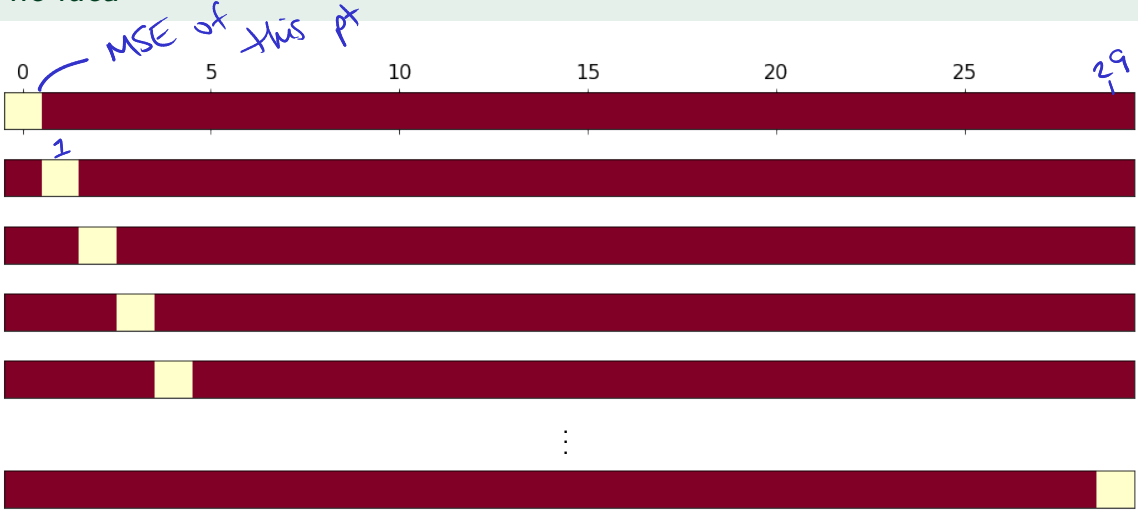
But Made training set smaller

Probably overestimating testing error

Section 3

Leave-One-Out Cross-Validation (LOOCV)

The idea



n=30

Repeat 30 times!

The idea in mathy words

- Remove (x_1, y_1) for testing.
- Train the model on $n-1$ points: $\{(x_2, y_2), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_1 = (y_1 - \hat{y}_1)^2$

- Remove (x_2, y_2) for testing.
- Train the model on $n-1$ points: $\{(x_1, y_1), (x_3, y_3), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_2 = (y_2 - \hat{y}_2)^2$

- Rinse and repeat

LOO-CV

Leave one out

Return the score:

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i$$

MSE when dropped (x_i, y_i)

LOOCV Pros and Cons

Advantages:

- Using almost all data all the time! 70% ⇝ n-1
- Stable results: No randomness so same # every time
- Evaluation is done w/ more testing data. 30% ⇝ all data

Disadvantages:

Runtime = Bad!

fitting model n times

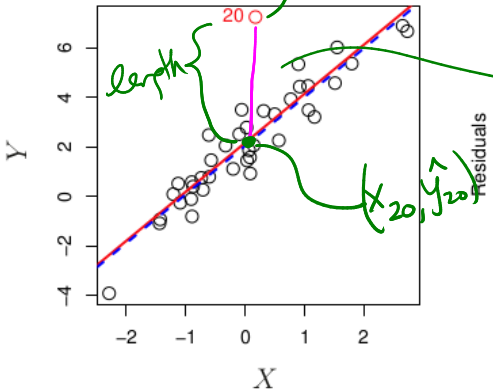
Goal: Cheap trick in special cases

Outliers

An *outlier* is a point for which y_i is far from the value predicted by the model.

(x_{20}, y_{20})

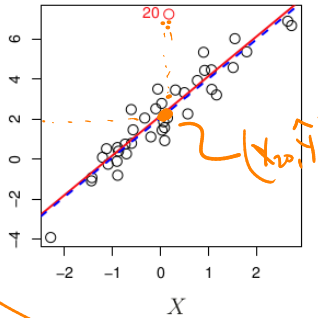
$|y_i - \hat{y}_i|$



Red solid: Regression w/ all data

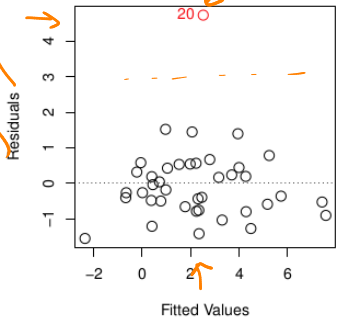
Blue dashed line without data pt 20

Residuals



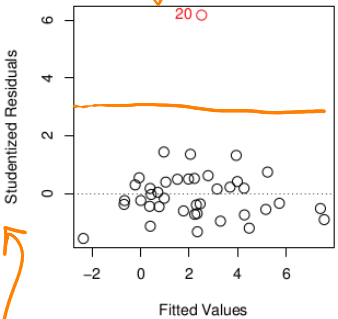
$(x_{20}, \hat{y}_{20})$

$y_i - \hat{y}_i$



(Y)

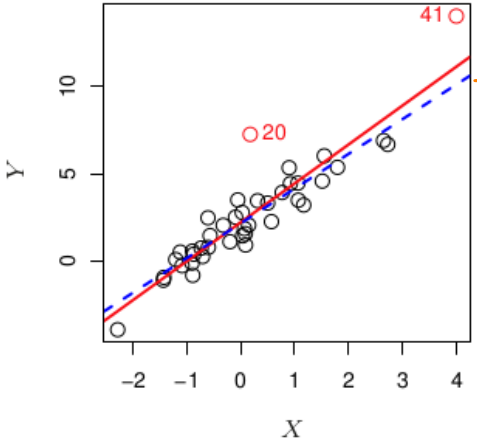
No clear way to pick cutoff



reasonable cutoff $=3$

divide by standard error

High Leverage



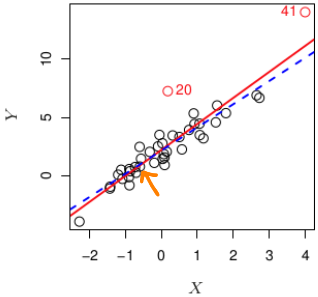
• Same data as before + pt 40

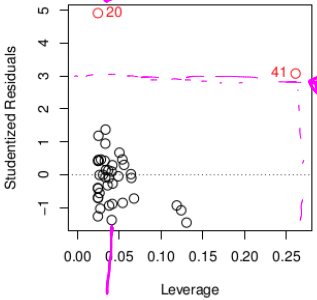
Red fit w/ all data

Blue dashed line fit without point 41

Observations with *high leverage* have an unusual value for x_i.

Leverage statistic





high residual (outlier)

low leverage (not high leverage)

$\frac{p+1}{n}$

Version for $p = 1$

$$h_i = \frac{1}{n} + \underbrace{\frac{(x_i - \bar{x})^2}{\sum_{j=1}^{n}(x_j - \bar{x})^2}}_{\text{same for all } i}$$

$X_1, \ldots, X_p$

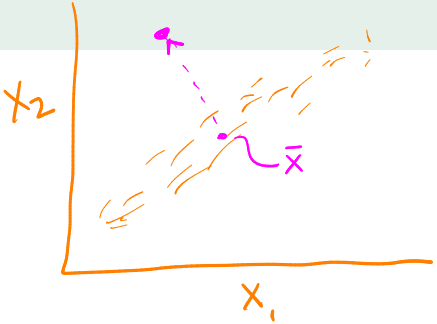
predict Y

Generalize to $p \geq 1$

But computer :)

Leverage statistic properties

$$h_i = \frac{1}{n} + \frac{(x_i - \overline{x})^2}{\sum_{j=1}^{n}(x_j - \overline{x})^2}$$



- Increases with distance from x_i to $\bar{x}$
- Always between $1/n$ and 1
- Avg h_i for all observations is $\frac{p+1}{n}$ $\quad \frac{2}{41} \approx 0.049$
- If $h_i \gg \frac{p+1}{n}$ high leverage

Speeding up LOOCV

Warning: This only works for least squares linear or polynomial regression.

already saw

computationally much cheaper

$$\frac{1}{n}\sum_{i=1}^{n}\mathrm{MSE}_i = CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_i}\right)^2$$

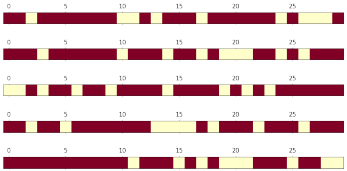
leverage

Build Lin Reg once on all data

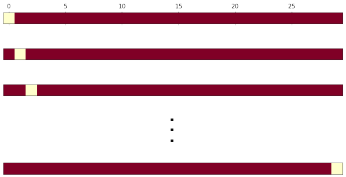
TL;DR

Any modeling I want

Validation set



LOO-CV



LOO-CV Score

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i$$

Cheap trick for regression

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_i}\right)^2$$

Next time

- Fri:
  - Lab on Classifiers
  - HW #4 due. Covers:
    - 2/4: Discrimination Lab
    - 2/2: 4.4.2, 4.4.3, 4.5 QDA, Naive Bayes, Some comparisons
    - 2/2: 5.1.1, 5.1.2 LOO Cross-Validation

Mon k-fold CV